

Nikita L. Polomoshnov

Computational drug design · machine learning · Moscow, Russia

nikitapol@fbb.msu.ru · github.com/doctawho42 · doctawho42.github.io · ORCID 0009-0001-4342-8539

Research interests. Physics-informed, interpretable machine learning: computing what physics or geometry fixes and learning only the residual, and measuring where a composed pipeline silently misleads, through unidentifiable quantities, uncalibrated uncertainty, and metrics whose winner changes with an unstated convention. Two of my manuscripts report pre-registered negative results as their central finding.

EDUCATION

Lomonosov Moscow State University, Faculty of Bioengineering & Bioinformatics

expected 2029

Specialist Degree (Специалист), 6-year programme; Master's-equivalent level, grants access to doctoral study. GPA 3.8/4.0.

- Second-prize diploma, XI and XII International Biological Universiade, Lomonosov MSU (2023 and 2024).

AIRI Summer School, Machine Learning track

2025

- R3-MoGAN: the R3GAN recipe (NeurIPS 2024) carried over to molecular graphs on QM9, with a latent-space variant on MOSES and an EDM diffusion baseline. github.com/g42alaxy/AIRI-Summer-School-Molecular-R3-GANs.

EXPERIENCE

Laboratory of AI in Genomics and Engineering Biology (AIGEN), Lomonosov MSU. Researcher

September 2026 – present

- Generative co-design of proteins and their guide RNAs in the TnpB/TrnC/Cas12 family: generating a nuclease for a given guide, and the inverse problem.

Yandex, Machine Learning Intern

October 2025 – February 2026

- Image-quality assessment: ranking and quality-classification models. Tag suggestion and SQL data preparation. Python, PyTorch, SQL.

Institute of Biomedical Chemistry (IBMC), Moscow. Researcher

2023 – present

- MetaTox on the [way2drug](https://way2drug.com) platform: metabolite-prediction service going into production. My contribution is a new prediction model and a new bank of transformation rules; the released bank and the code are public at github.com/doctawho42/GRAIL.
- Enantiomer screening campaign on molecular cages with the Novosibirsk Institute of Organic Chemistry (NIOCH SB RAS): docking and target scoring, with an operational report separating enantiomers on one target.

PUBLICATIONS & MANUSCRIPTS

*Author order reflects contribution; * denotes sole authorship.*

What Cycle Closure Can and Cannot See in a Relative Binding Free Energy Network *

N. L. Polomoshnov, sole author · preprint on ChemRxiv

- The per-edge BAR uncertainty, calibrated against replicate scatter on 1145 public OpenFE binding edges, becomes the null of a cycle-closure goodness-of-fit test; a gradient-cycle decomposition fixes what such a test can and cannot see. doi.org/10.26434/chemrxiv.15007931/v3.

Ground Truth That Does Not Ground: substituting reference σ -profiles degrades COSMO-SAC solubility prediction

N. L. Polomoshnov, A. V. Rudik (IBMC) · under review, J. Chem. Inf. Model.

- A differentiable COSMO-SAC layer lets a graph network train through a fixed thermodynamic map, and an exact decomposition separates the map's misspecification from what its inputs fail to carry. Substituting the tabulated profile at inference degrades prediction at every seed; supervising against it during training carries no established sign. doctawho42.github.io/tgmn-solv

Metabolite Predictors Have a Multiverse Problem

N. L. Polomoshnov, A. V. Rudik (IBMC) · manuscript, J. Chem. Inf. Model.

- recall@k factorises exactly into coverage × selection × ranking: on the comparison set the bank ceiling is 0.8105 and the system reaches 0.7023, so the remaining 0.1895 is out of reach of any ranking. The comparison against four published predictors is reported as a grid, since varying the matching criterion moves the verdict at 6 of 9 output budgets. Code and rule bank: github.com/doctawho42/GRAIL.

Machine-learning evaluation methodology *

N. L. Polomoshnov, sole author · under review, ICLR 2027

- Title and details withheld during double-blind review.

SELECTED OPEN-SOURCE & COMMUNITY

- OpenFreeEnergy / IndustryBenchmarks2024: issue [#331](#) (per-edge observability audit with reproducible tables and a 33-check self-test) and issue [#332](#) (ligand-name collision in the cdk8 set). Entrant, OpenADMET CYP Inhibition Blind Challenge 2026.

TECHNICAL SKILLS

Machine learning: PyTorch; graph neural networks; E(3)-equivariant architectures; generative models on graphs (GAN, diffusion); deep ensembles; calibration & uncertainty quantification; conformal prediction; GFlowNets.

Simulation & cheminformatics: molecular dynamics & alchemical free-energy calculations (OpenFE / OpenFF, OpenMM); MBAR/BAR (pymbar); COSMO-SAC / COSMO-RS; RDKit; docking.

Bioinformatics: sequence analysis and homology search (BLAST); sequence and structure databases; NGS data processing; structural bioinformatics (PyMOL, GROMACS).

Statistics: M-estimation & sandwich variance; paired and clustered bootstrap; multiplicity; pre-registration; negative controls.

Engineering: reproducible pipelines; artifact / DOI release; Git, Docker, SQL.

Languages: Russian (native). English (professional working proficiency; qualified in scientific-text review and translation).